# Single Photon Microscopy

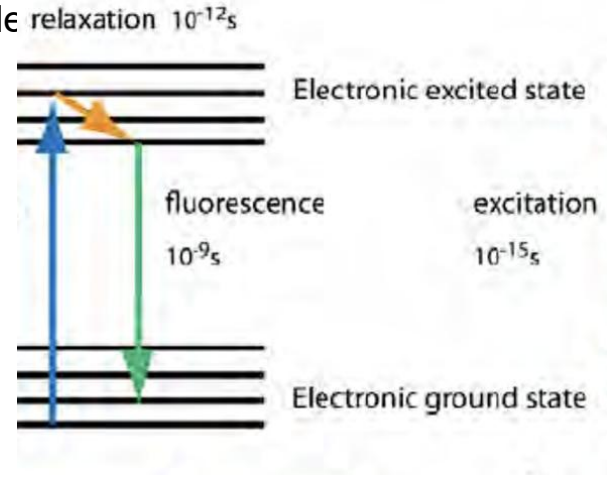
Only including section created by Avery Books
This was made for EE218A @ UC Berkeley
with Prof. Kante
Speaker Notes: Notion

Berkeley
UNIVERSITY OF CALIFORNIA

# How Single Photon Microscopy Works

**What is Single Photon Microscopy?**

Uses an external light source to **pump** specimen that is embedded with **single-photon emitters** that absorb a photon from the pump, excite and de excite it's electron, and emit single photons which are detected by the microscope array.

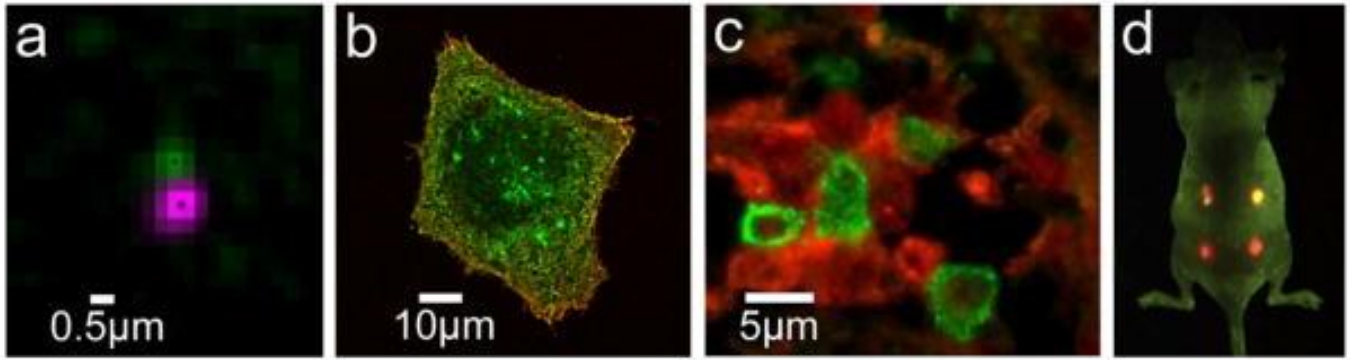



- Single Photon Sources - Quantum Dots, Organic Dyes, Fluorescent Proteins
- Engineering of single photon sources
- Applications of single photon
- Brief comparison to another method

Source: [14]

# Single Photon Sources: Why Important?

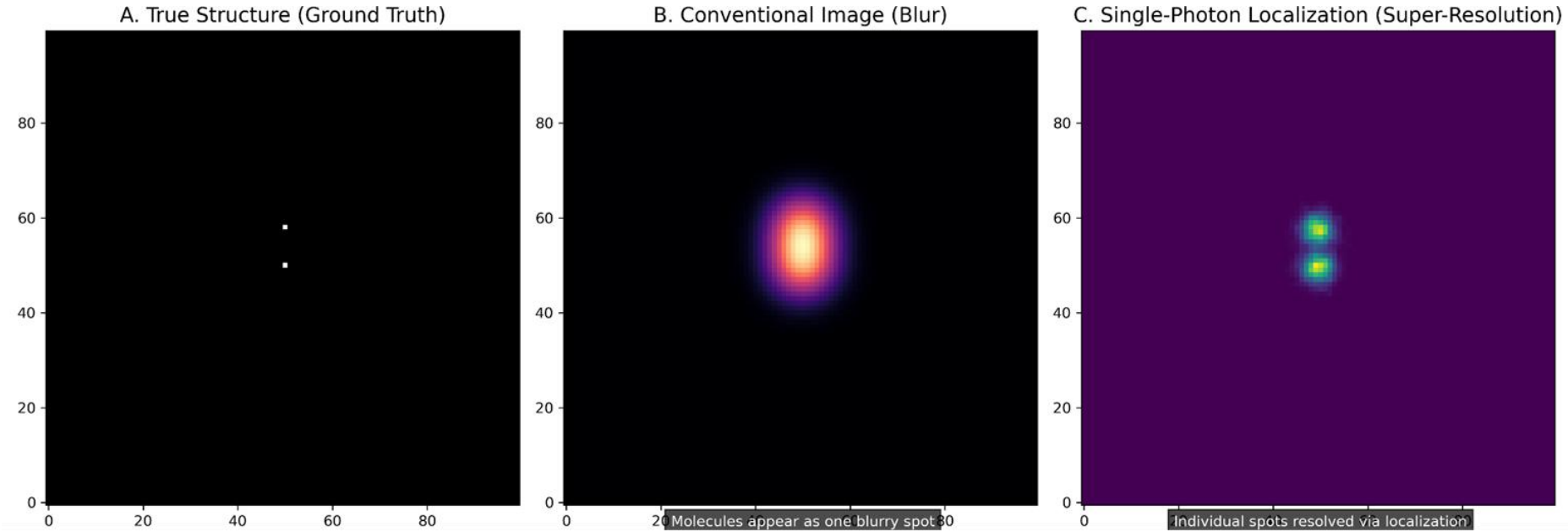
1. Break resolution limit by isolation of PSFs
2. Sources integrated into object




| | Single Molecule | Single Cell | Tissue | In vivo |
|---|---|---|---|---|
| **Modality** | widefield<br>confocal<br>TIRF | widefield<br>confocal<br>TIRF | widefield<br>confocal<br>spectral imaging | whole animal<br>intravital<br>two-photon |
| **Spatial resolution** | nm | μm | μm | μm-mm |
| **Temporal resolution** | ms | ms-min | N/A | s-hr |

[2] Vu, T. Q., et al. (2015). Quantum dots for quantitative ɔved Single-imaging: from single molecules to tissue.

# Python Simulation - Comparison with Std. Microscopy



# Single Photon Sources: Types

Quantum Dots (QDs)

0-dimensional nanoscale semiconductor crystal that “blinks” and behaves as an artificial atom

Made via “wet chemistry”

0-dimensional

- Electrons spatially constricted s.t. <u>0 d.o.f.</u>
- Forces electron to <u>discrete</u> energy levels

“Blinking”

- Inherent <u>stochastic binary</u> states of fluorescence
- Sparse random subset of probes in the dot are “on” at any given point and are required to <u>locally isolate</u>

# Single Photon Sources: Types

Organic Dyes

Small Molecule dyes that are conjugated to antibodies or other agents that blink due to chemical buffers

Chemical Buffers → Requires optimized and cytotoxic imaging buffers to induce blinking

Optical properties → Less photo stable and less total photons than QDs before photobleaching

Optical properties → Small size minimizes linkage errors -> better map of location

# Single Photon Sources: Types

Fluorescent Proteins

Genetically encoded labels that are produced by cell

Photoactivatable → Exist in “off” state and turn “on” when excited (irreversible)

Photoconvertible → Change emission color after activation by particular wavelength

Properties → Endogenous and biocompatible

Properties → Low photon count and photostability

Fig from:

Interventions for preventing or controlling health care–associated infection among health care workers or patients within primary care facilities: A scoping review

Lucyna Gozdzielewska, PhD, et. al.

| | QDs | Organic Dyes | Fluorescent Proteins |
|---|---|---|---|
| Size | 1 | | |
| Photostability | | 2 | 2 |
| Broad excitation | 3 | | |
| Multiplex imaging | 4 | | |
| Brightness | 5 | | 6 |
| Flexible conjugation | 7 | | |
| Genetically expressible | 8 | 8 | |
| Electron dense | | 9 | 9 |
| Sample fixation requirements | 10 | | |
| Continuous emission | 11 | | |
| Accessible time/length scales | 12 | 13 | 13 |
| In vivo imaging | 14 | | 15 |

# Spontaneous Emission

Quantum Mechanics & Electrodynamics govern the emission parameters:

Dirac Theory of Radiation → Transition of states is viewed as interaction with vacuum fluctuations of QEM field (random phase)

Fluorescence Parameters →
- Molar Extinction Coefficient (probability to absorb)
- Fluorescence Quantum Yield (brightness)
- Fluorescence Lifetime (avg. time in excited state)

# Comparison to Differential Interference Contrast (DIC) Microscopy

Single Photon:

1. Incoherent Fluorescence
2. Chemical Map: shows precise location of specific labeled molecules
3. Intensity
4. Limited by Diffraction (not as much)

Differential Interference Contrast

1. Phase shifts cause by sample's refractive index
2. Shows surface contours and optical density
3. Phase converted to pseudo - 3D shadow cast images
4. Limited by diffraction

# Sources


1. Urban, J. M., Chiang, W., Hammond, J. W., Cogan, N. M. B., Litzburg, A., Burke, R., Stern, H. A., Gelbard, H. A., Nilsson, B. L., & Krauss, T. D. (2021). Quantum Dots for Improved Single-Molecule Localization Microscopy. *The journal of physical chemistry. B*, *125*(10), 2566–2576. https://doi.org/10.1021/acs.jpcb.0c11545
2. Vu, T. Q., Lam, W. Y., Hatch, E. W., & Lidke, D. S. (2015). Quantum dots for quantitative imaging: from single molecules to tissue. *Cell and tissue research*, *360*(1), 71–86. https://doi.org/10.1007/s00441-014-2087-2
3. Betzig, E., Patterson, G. H., Sougrat, R., Lindwasser, O.W., Olenych, S., Bonifacino, J. S., Davidson, M. W., Lippincott-schwartz, J., Hess, H. F. (2006). Imaging Intracellular Fluorescent Proteins at Nanometer Resolution. *Science*. Imaging Intracellular Fluorescent Proteins at Nanometer Resolution
4. Zhong, Haining. (2010). Photoactivated Localization Microscopy (PALM): An Optical Technique for Achieving 10-nm Resolution. Cold Spring Harbor protocols. 2010. pdb.top91. 10.1101/pdb.top91.Photoactivated Localization Microscopy (PALM): An Optical Technique for Achieving 10nm Resolution
5. Simply Forensics. (2025). Unveiling the Microscopic World: A Comprehensive Guide to Microscopy Techniques. *Simplyforensic*. Microscopy Techniques: A Complete Guide to Imaging Methods
6. iBiology Science Stories. (2011, Jan 8). Eric Betzig and Harald Hess (Janelia Farm/HHMI): Developing PALM Microscopy https://www.youtube.com/watch?v=GcQ24khZzvU&t=558s
7. "History of Microscopes." *Microscope.com*, Education Center, Microscope.com, www.microscope.com/education-center/articles/history-of-microscopes. Accessed 24 Nov. 2025.
8. Dahal, Prashant. *"Microscopy: History, Classification, and Terms."* Microbe Notes, 24 Nov. 2023, microbenotes.com/microscope/.
9. "Microscope Optical Components Introduction." *Evident Scientific – Knowledge Hub*, Evident Scientific, www.evidentscientific.com/en/microscope-resource/knowledge-hub/anatomy/components. Accessed 24 Nov. 2025.
10. Betzig, E., Patterson, G. H., Shroff, H., Lippincott-schwartz, J., Davidson, M. W., "Introduction to Photoactivated Localization Microscopy" *Zeiss*. Zeiss: Introduction to PALM
11. Murphy, D. B., Moulding, R. S., Spring, K. R., Schwartz, S., & Davidson, M. W. (n.d.). Introduction to Phase Contrast Microscopy. Nikon's MicroscopyU. https://www.microscopyu.com/techniques/phase-contrast/introduction-to-phase-contrast-microscopy
12. Chen, N., Gao, G., Chong, S. P. (2012)."Focal Modulation Microscopy: Principle and Techniques" *National University of Singapore*. 10.5772/30906. Focal Modulation Microscopy
13. Dirac, P. A. M. (1927). The Quantum Theory of the Emission and Absorption of Radiation. Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character, 114(767), 243–265